



Communications and Networks

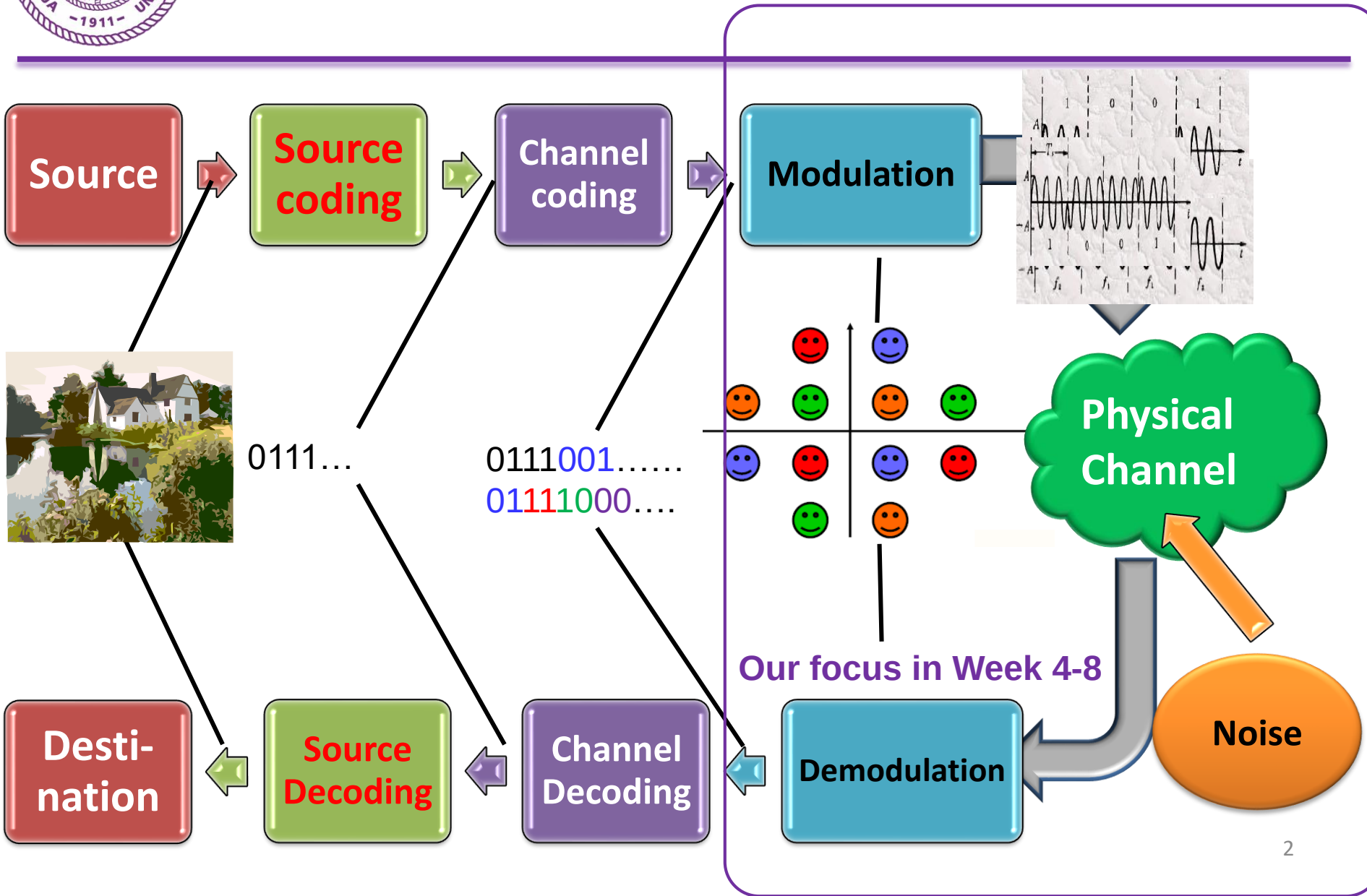
Lecture 4: Baseband Transmission

- Optimal Receiver

Sheng Zhou

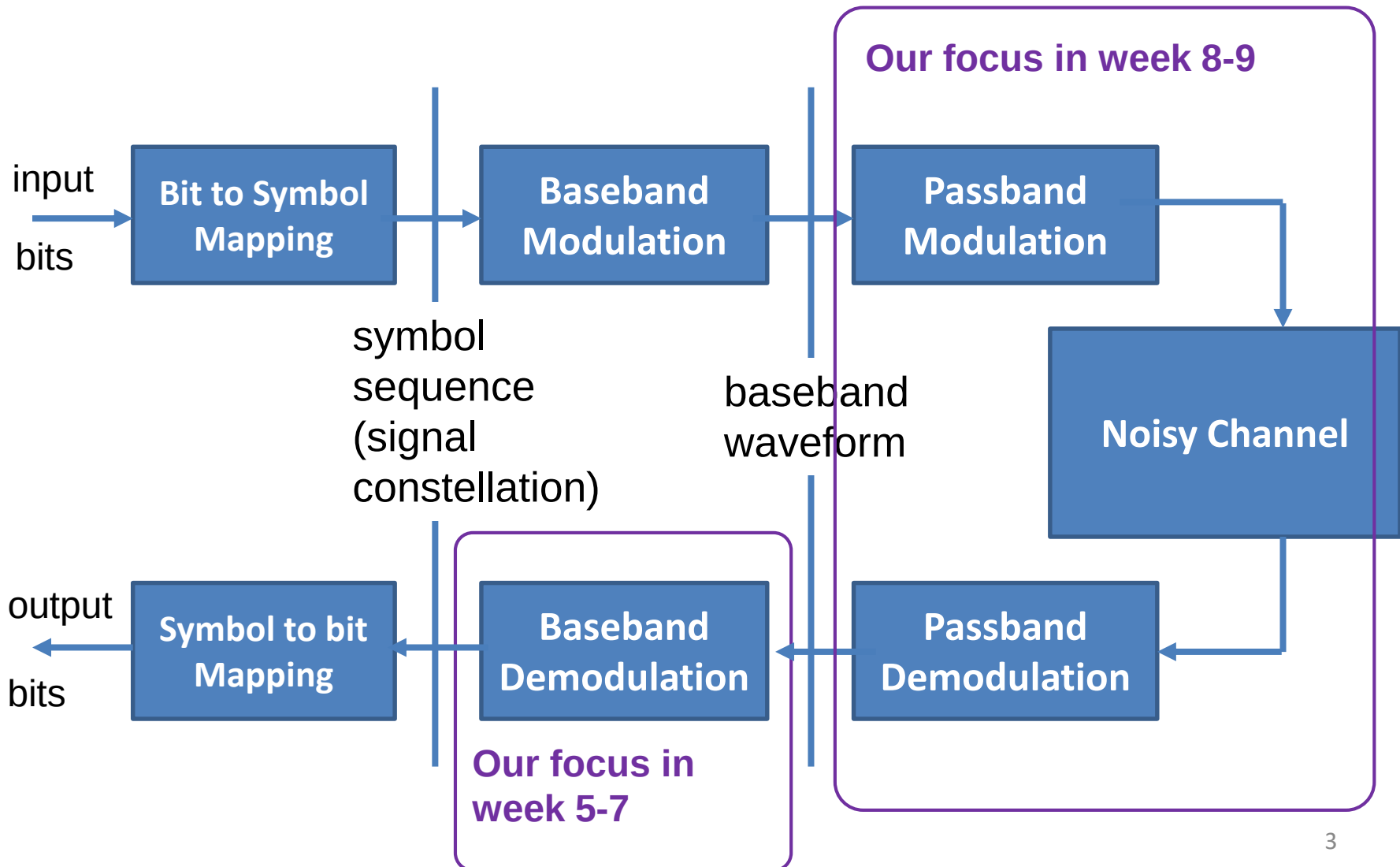


Digital Communication Link: **Building Blocks**



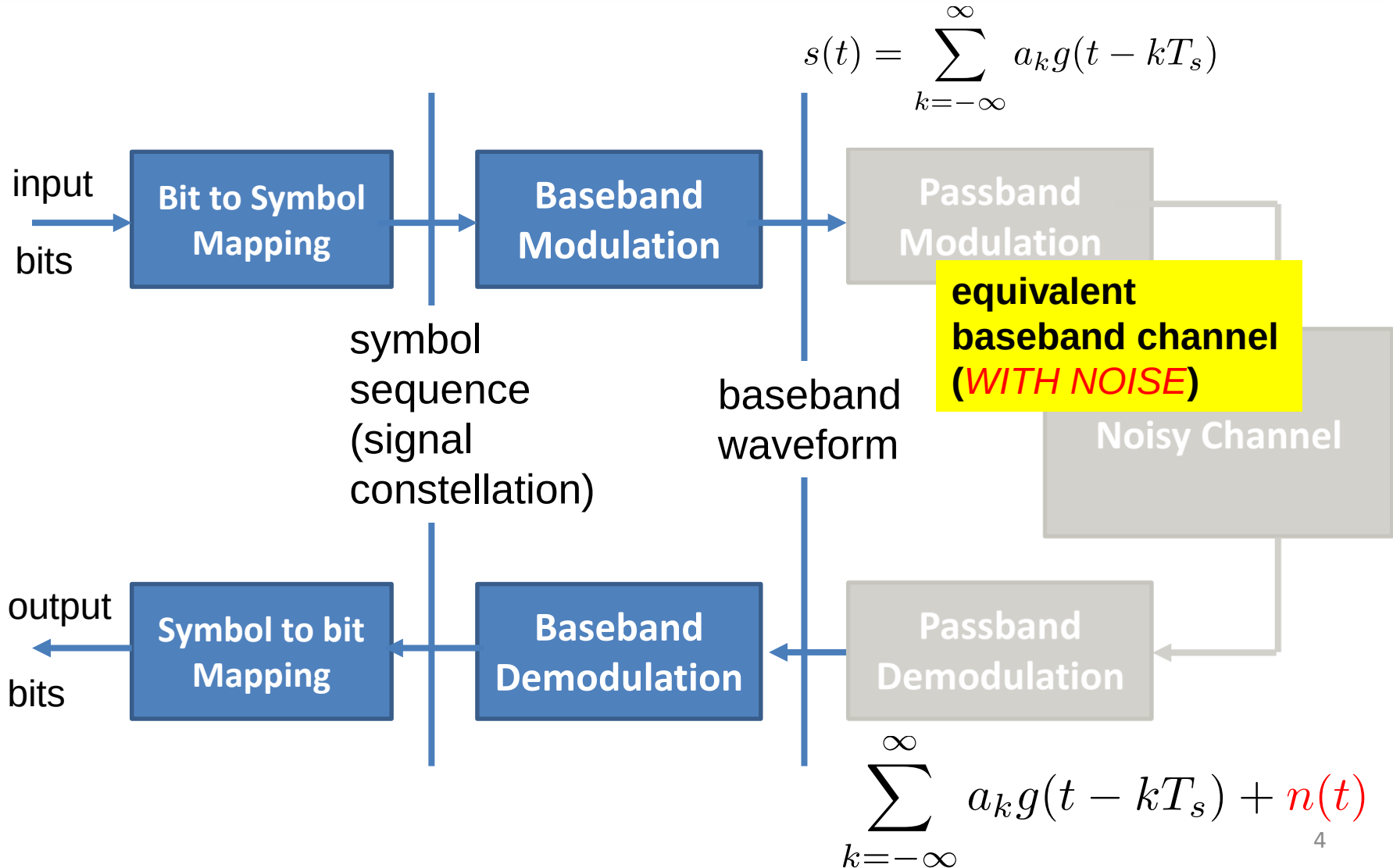


Modulation: Overview





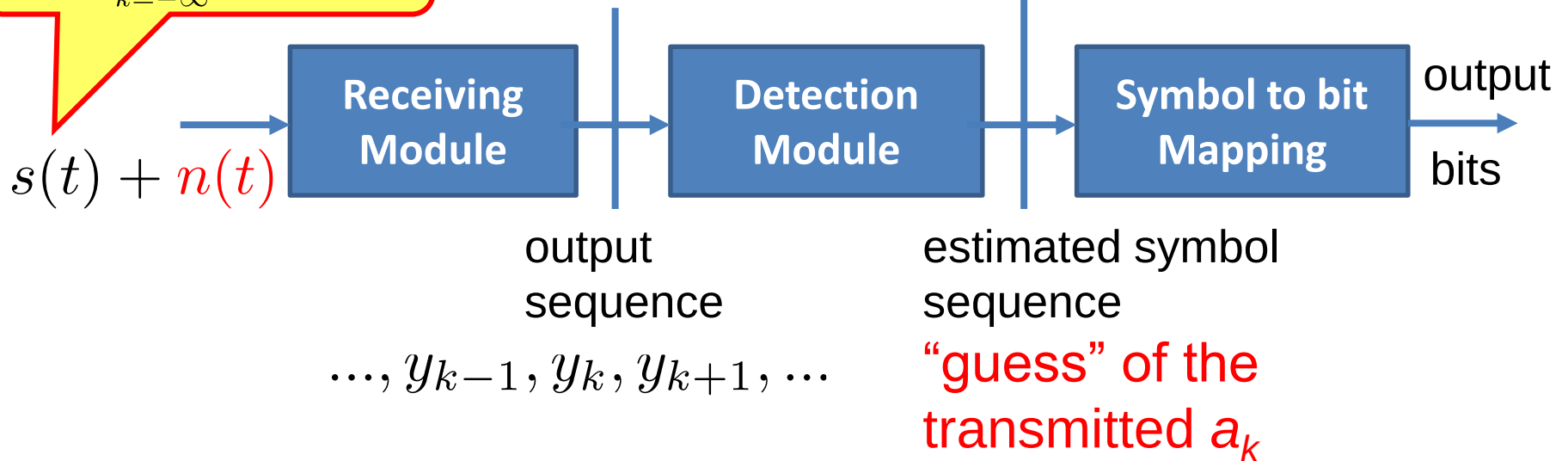
Modulation: Overview





Baseband Demodulator: Components

$$s(t) = \sum_{k=-\infty}^{\infty} a_k g(t - kT_s)$$



- Property of the noise (this lecture)
- How to design the receiving module (this lecture)
- How to make the symbol decision (next lecture)



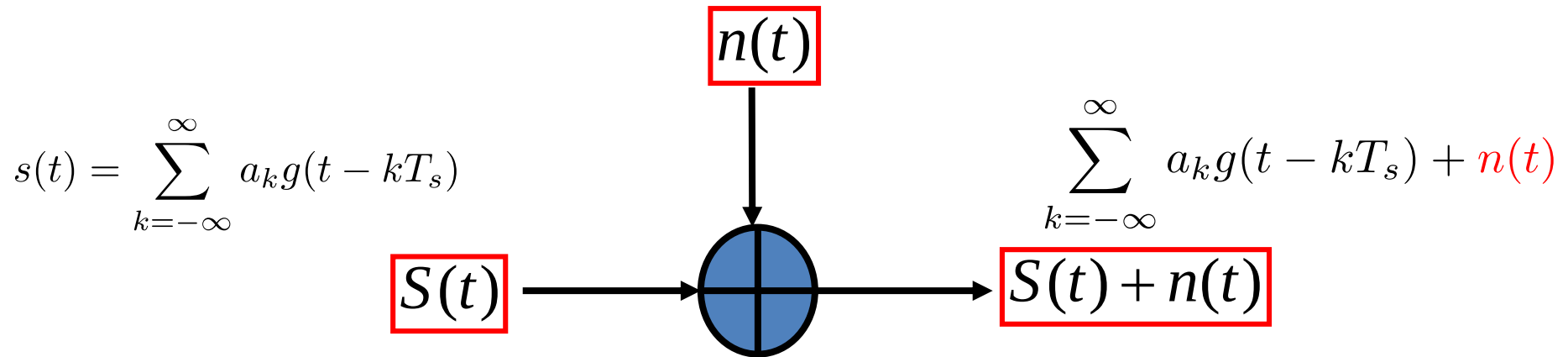
Outline

- Additive Gaussian Noise
- Matched Filter
- Signal to Noise Ratio



Baseband Transmission Model

■ Baseband channel **with additive noise** :



- ▶ This course only consider additive noise
 - Thermal noise from the channel and the devices
- ▶ The noise is a **random process**
- ▶ Due to noise, it is not an easy job to guess a_k from $S(t)+n(t)$



Noise as a **Gaussian Process**

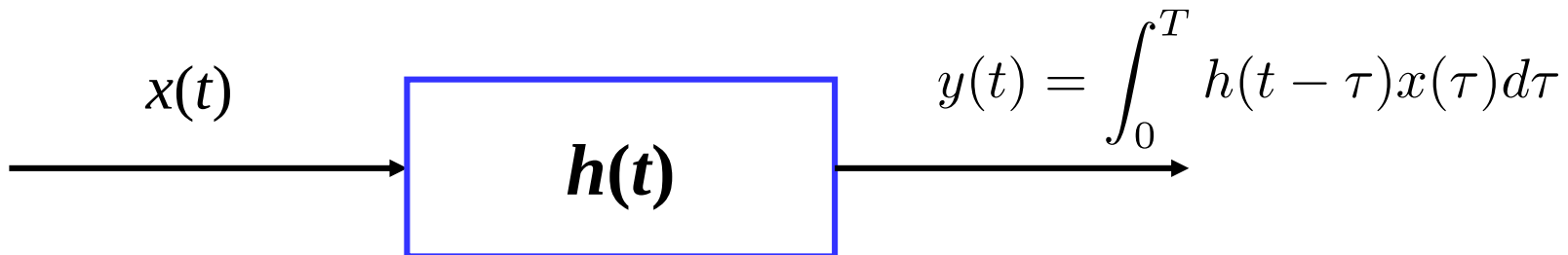
$$Y = \int_0^T g(t)x(t)dt$$

- For any $g(t)$ that makes Y being a random variable with finite variance, if Y is a Gaussian variable, $x(t)$ is a **Gaussian Process**

$$f_Y(y) = \frac{1}{\sqrt{2\pi}\sigma_Y} \exp \left[-\frac{(y - \mu_Y)^2}{2\sigma_Y^2} \right]$$

► Hint: **linear combination** of (joint) gaussian variables are Gaussian

- Useful property: A Gaussian process passes a LTI system, the output process $y(t)$ is still a Gaussian process





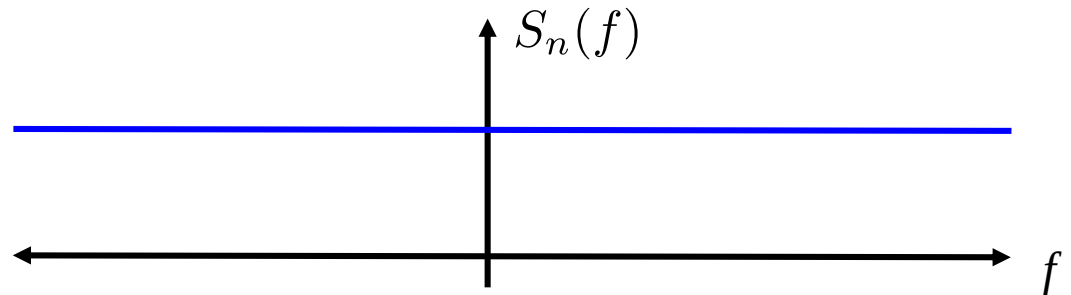
White Gaussian Noise

- We general consider *white* Gaussian noise $n(t)$
- The autocorrelation function of $n(t)$ is

$$R_n(\tau) = \frac{n_0}{2} \delta(\tau)$$

- Then the power spectrum density (PSD) of $n(t)$ is

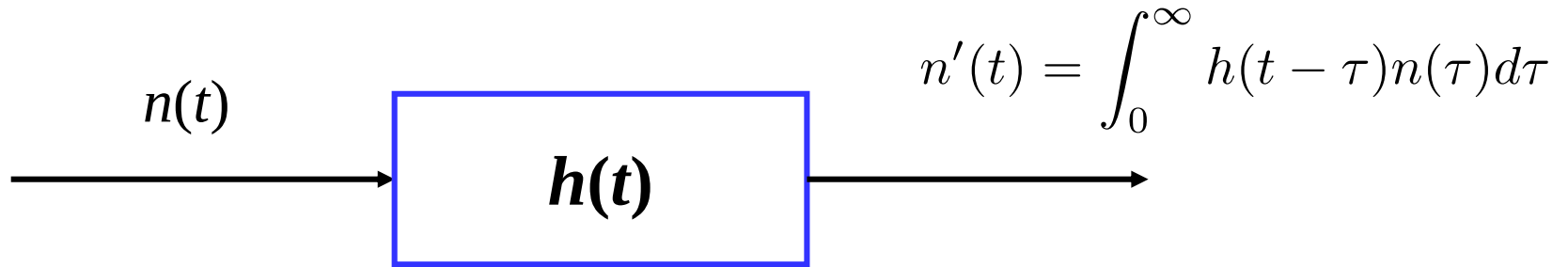
$$S_n(f) = \frac{n_0}{2}$$



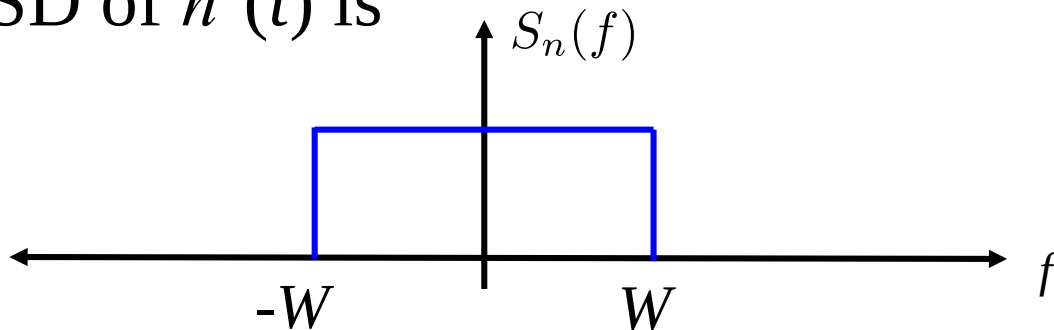
- PSD has all the frequency components: White!
- What is the average power of white Gaussian Noise?



Low-Pass Filtered White Gaussian Noise



- If $H(f)$ is an ideal low-pass filter with bandwidth W
- The PSD of $n'(t)$ is



- What is the average power of $n'(t)$?

► $n_0 W$



Recall Nyquist Criterion and Shannon Formula

- Nyquist Criterion: $R_s \leq 2W$
- The capacity of the AWGN baseband channel is

$$C = \frac{1}{2} \log \left(1 + \frac{S}{Wn_0} \right) \times 2W = W \log \left(1 + \frac{S}{Wn_0} \right)$$

- Here we should use the ideal low-pass filter as the baseband pulse
- Observe: Need to maximize the SNR
- Let's see how to design receivers



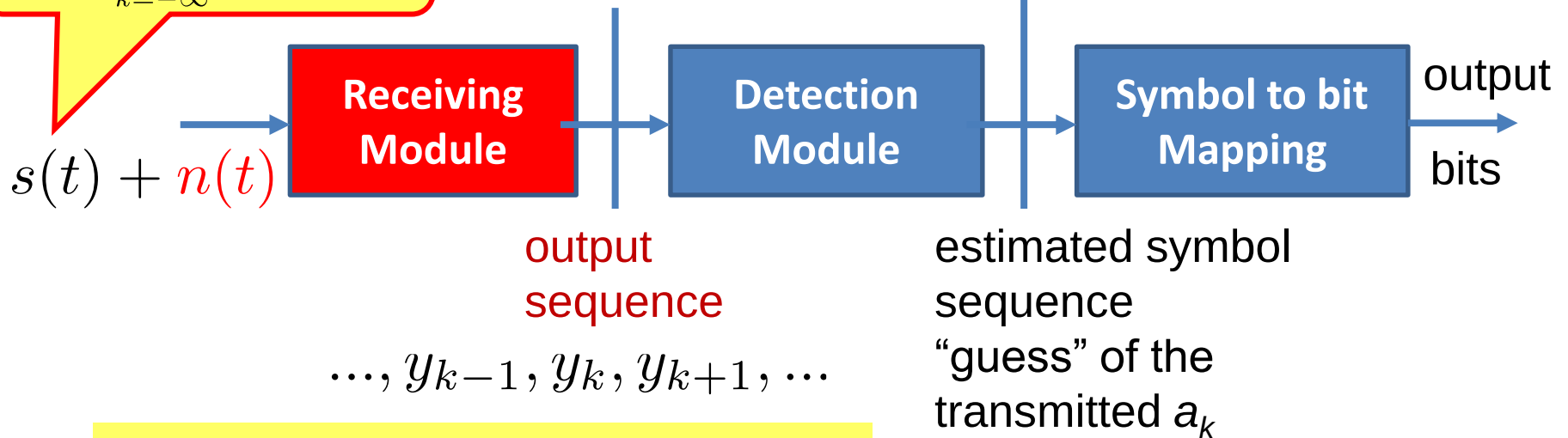
Outline

- Additive Gaussian Noise
- Matched Filter
- About Signal to Noise Ratio



Baseband Demodulator: Components

$$s(t) = \sum_{k=-\infty}^{\infty} a_k g(t - kT_s)$$



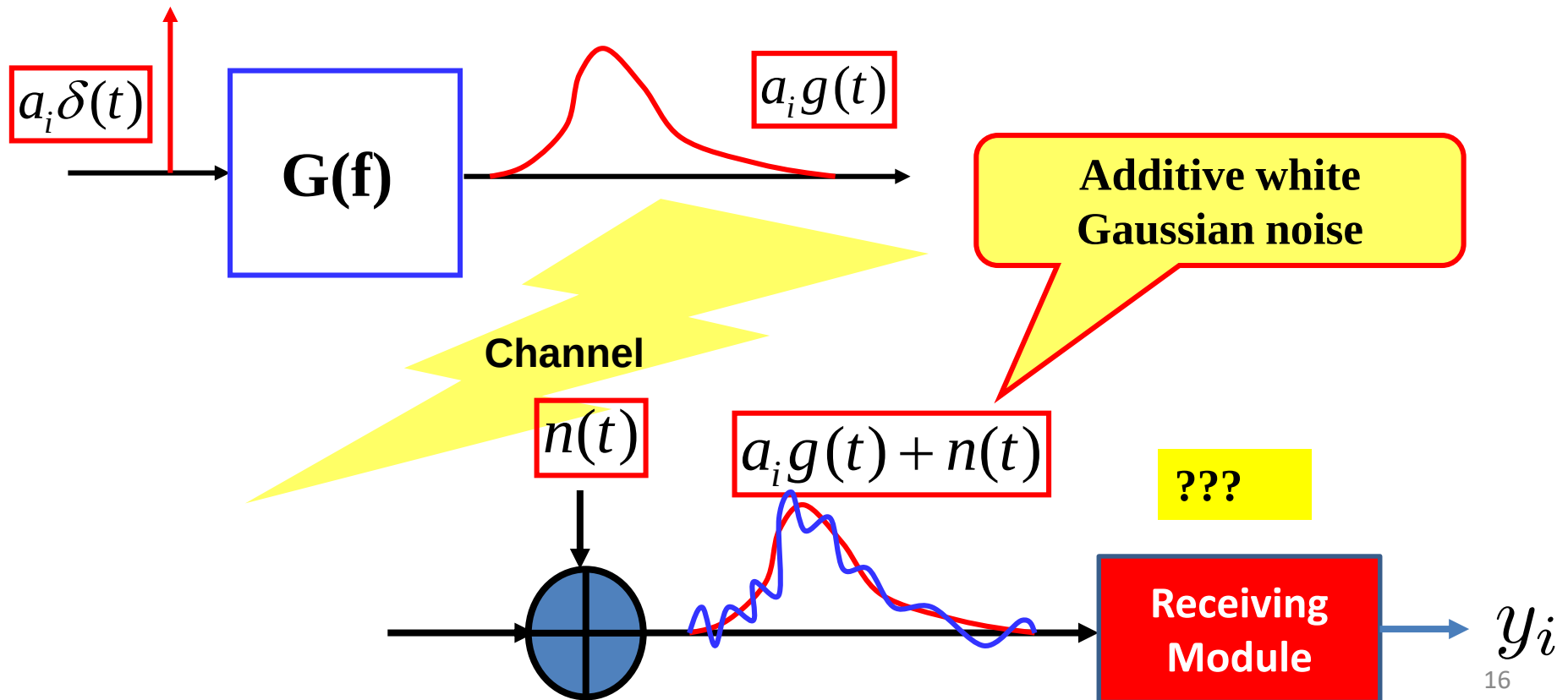
Goal: Maximize the SNR of y_k

- Property of the noise (this lecture)
- How to design the receiving module (this lecture)
- How to make the symbol decision (next lecture)



Let's focus on **one symbol**

- Consider the waveform corresponding to one symbol
 - What about inter symbol interference (ISI) ?





Receiver Architecture Option 1

$$s(t) = \sum_{k=-\infty}^{\infty} a_k g(t - kT_s)$$

symbol
sequence
(**signal
constellation**)

Baseband
Modulation

Passband
Modulation

**equivalent
baseband channel
(**WITH NOISE**)**

Noisy Channel

Option 1:
Sample at mT_s , n is an integer
No-ISI due to Nyquist Criterion

baseband
waveform

Detection and
Decision

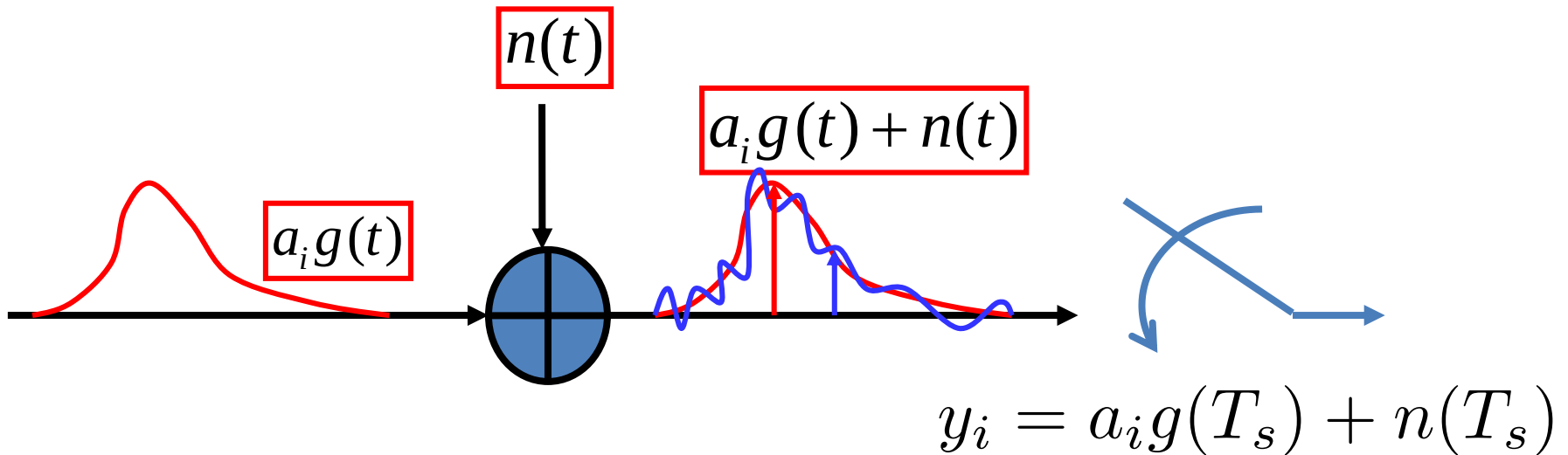
Passband
Demodulation

$$y_m = a_m g(0) + \textcolor{red}{n}(mT_s)$$

$$\sum_{k=-\infty}^{\infty} a_k g(t - kT_s) + \textcolor{red}{n}(t)$$



Direct Sample at T_s

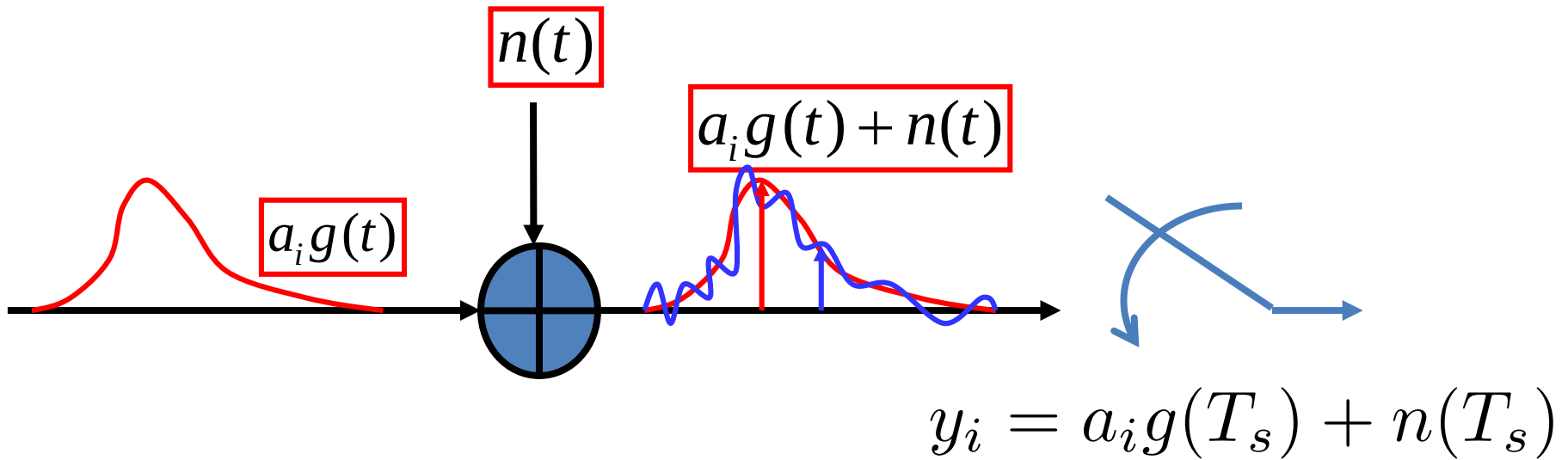


Suppose we sample at $t=T_s$ (for causality reasons, the waveform $g(t)$ itself is also shifted by T_s , so **we assume its peak is also at T_s**)

How does it perform ??



SNR of the Sampling Output



$$\text{SNR} = \frac{|a_i g(T_s)|^2}{\mathbb{E}[n^2(T_s)]} = 0$$

SNR is too low!



Receiver Architecture **Option 2**

$$s(t) = \sum_{k=-\infty}^{\infty} a_k g(t - kT_s)$$

symbol
sequence
(**signal
constellation**)

Baseband
Modulation

Passband
Modulation

**equivalent
baseband channel
(*WITH NOISE*)**

baseband
waveform

Noisy Channel

**Option 2:
Add a filter before the sampler**

Detection and
Decision

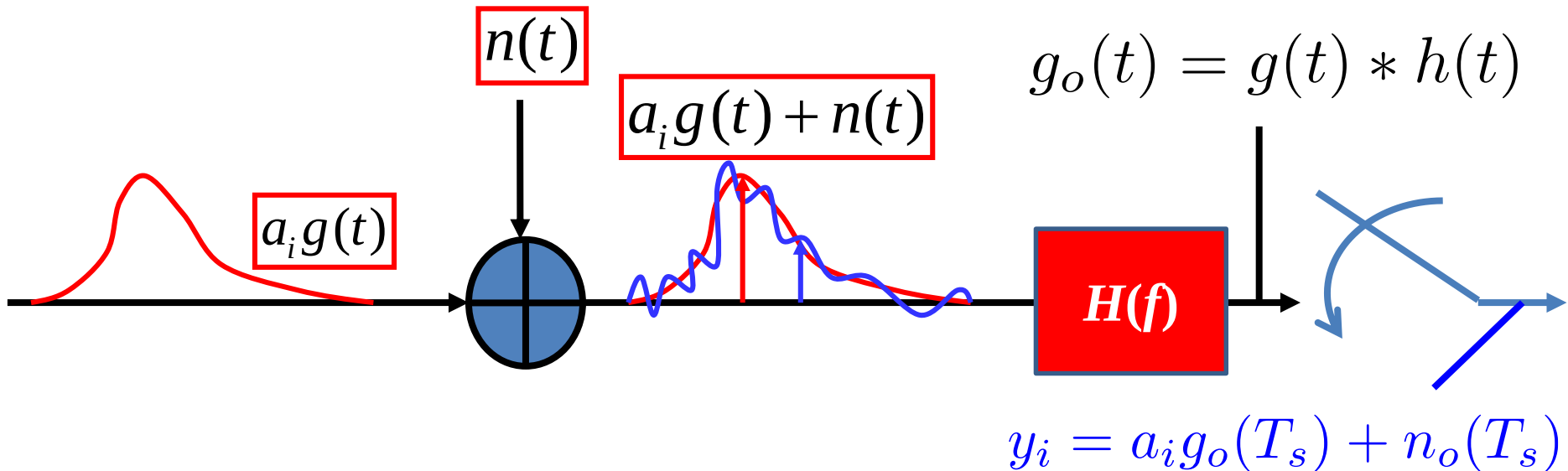
$H(f)$

Passband
Demodulation

$$\sum_{k=-\infty}^{\infty} a_k g(t - kT_s) + \underset{20}{n}(t)$$

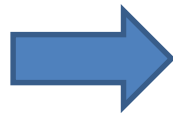


SNR of the Sampling Output after a Filter



$$g_o(t) = g(t) * h(t)$$

$$n_o(t) = n(t) * h(t)$$

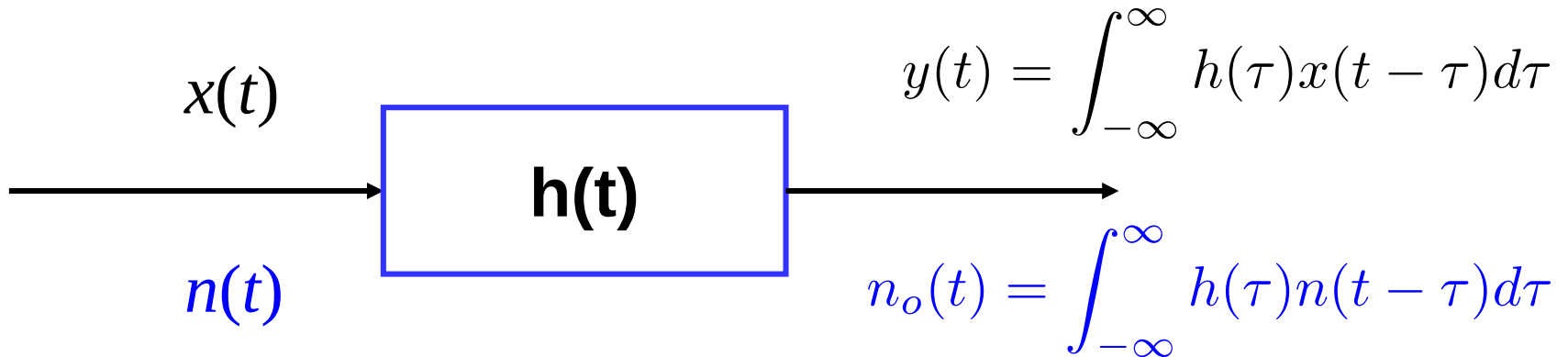


$$\text{SNR} = \frac{|a_i g_o(T_s)|^2}{\mathbb{E}[n_o^2(T_s)]}$$

Maximize SNR by optimizing filter $h(t)$



Noise Power after the Filter



- Recall: Power Spectrum of a Random Process Passes through a Filter:

$$S_Y(f) = |H(f)|^2 S_X(f)$$

- Now for the white Gaussian noise: $S_{N_o}(f) = \frac{n_0}{2} |H(f)|^2$

➡ $\mathbb{E}[n_o^2(T_s)] = R_{N_o}(0) = \int_{-\infty}^{\infty} S_{N_o}(f) e^{j2\pi f \cdot 0} df = \int_{-\infty}^{\infty} \frac{n_0}{2} |H(f)|^2 df$



Signal Power at the output sample point

- Frequency response of the output signal:

$$G_o(f) = G(f)H(f)$$

- Take inverse Fourier transform

$$g_o(T_s) = \int_{-\infty}^{\infty} G(f)H(f)e^{j2\pi fT_s} df$$

- The signal power is then

$$|a_i g_o(T_s)|^2 = \left| a_i \int_{-\infty}^{\infty} G(f)H(f)e^{j2\pi fT_s} df \right|^2$$



Maximizing SNR at the output sample point

$$\text{SNR} = \frac{\left| a_i \int_{-\infty}^{\infty} G(f) H(f) e^{j2\pi f T_s} df \right|^2}{\frac{n_0}{2} \int_{-\infty}^{\infty} |H(f)|^2 df}$$

How to maximize SNR by choosing H(f) ?

$$\left| \int_{-\infty}^{\infty} G(f) H(f) e^{j2\pi f T_s} df \right|^2 \leq \int_{-\infty}^{\infty} |G(f)|^2 df \int_{-\infty}^{\infty} |H(f)|^2 df$$

Equality holds when $H(f) = bG^*(f)e^{-j2\pi f T_s}$

*We have that g(t) has finite energy, and here b is a constant



Maximum SNR at the output sample point

$$\text{SNR} = \frac{\left| a_i \int_{-\infty}^{\infty} G(f) H(f) e^{j2\pi f T_s} df \right|^2}{\frac{n_0}{2} \int_{-\infty}^{\infty} |H(f)|^2 df}$$

$$H(f) = b G^*(f) e^{-j2\pi f T_s}$$

■ Therefore the maximum SNR:

$$\text{SNR}_{\max} = \frac{\left| a_i \int_{-\infty}^{\infty} b |G(f)|^2 df \right|^2}{\frac{n_0}{2} \int_{-\infty}^{\infty} b^2 |G(f)|^2 df} = \frac{2}{n_0} |a_i|^2 \int_{-\infty}^{\infty} |G(f)|^2 df$$



Time Domain of Matched Filter

$$h(t) = g(T_s - t)$$

■ Let's revisit the convolution $g_o(t) = g(t) * h(t)$

$$\begin{aligned} g_o(t) = g(t) * h(t) &= \int_{-\infty}^{\infty} g(\tau) h(t - \tau) d\tau \\ &= \int_{-\infty}^{\infty} g(\tau) g(\tau + T_s - t) d\tau \end{aligned}$$

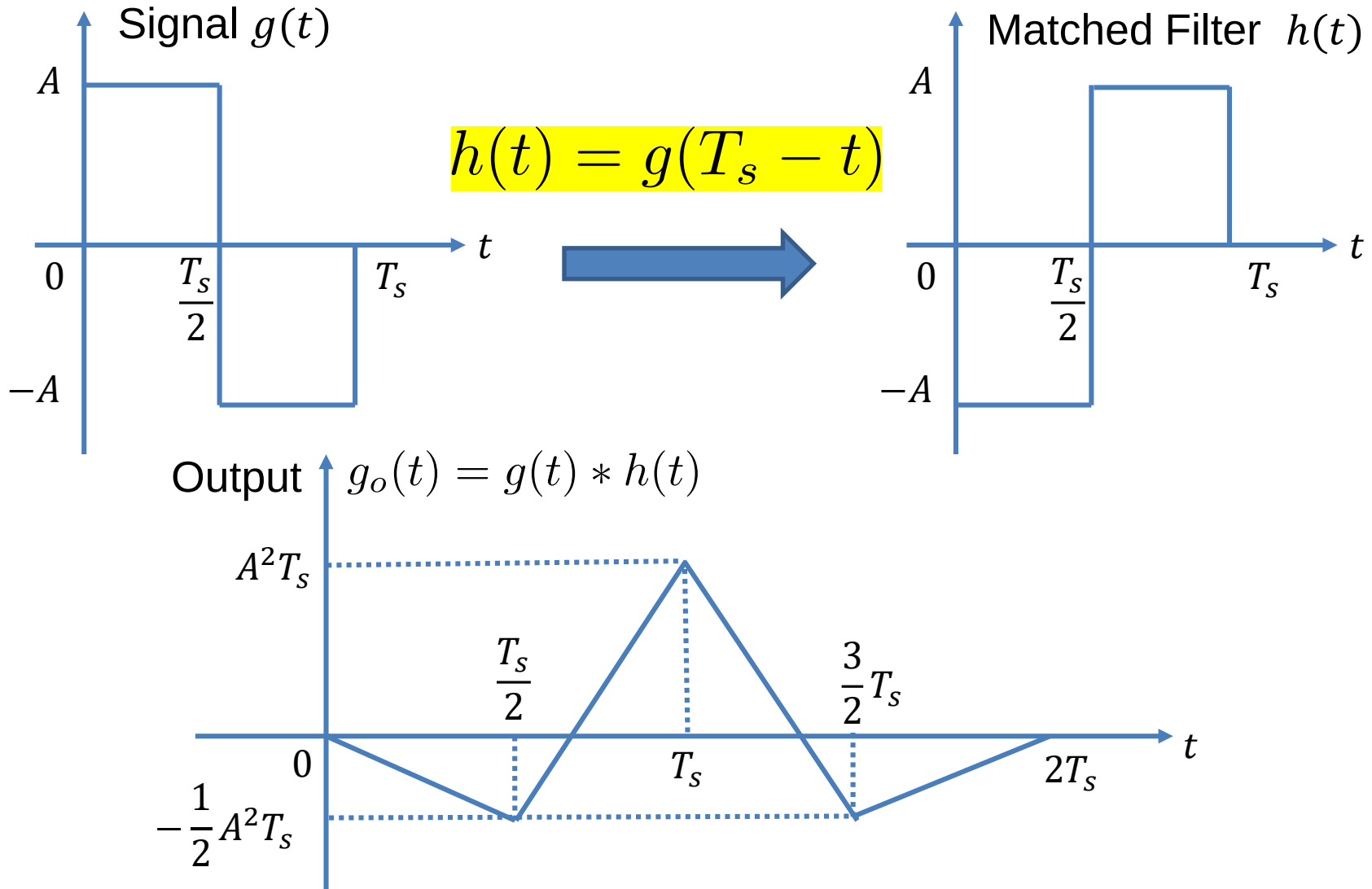
$$g_o(T_s) = \int_{-\infty}^{\infty} |g(\tau)|^2 d\tau$$

Equivalent to perfectly aligned *correlation*!

Time domain derivation of $h(t)$ is left for exercise



Example of a Matched Filter

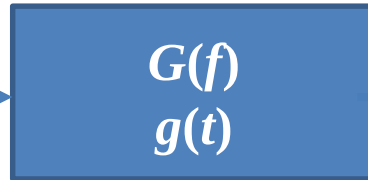




Revisit the Receive Architecture

$$s(t) = \sum_{k=-\infty}^{\infty} a_k g(t - kT_s)$$

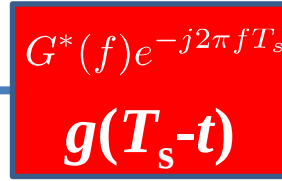
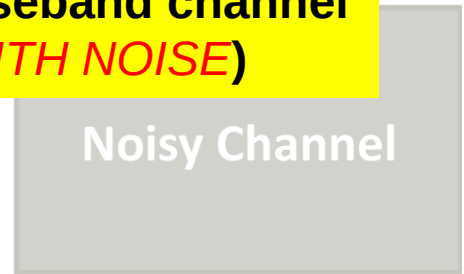
symbol
sequence
(**signal
constellation**)



**equivalent
baseband channel
(**WITH NOISE**)**

baseband
waveform

**Add a filter before the sampler
Are we free of ISI after the
sampler ?????**



$$y_m = \sum_{k=-\infty}^{\infty} a_k g_o(mT_s - kT_s) + \textcolor{red}{n}_o(mT_s)$$

$$\sum_{k=-\infty}^{\infty} a_k g(t - kT_s) + \textcolor{red}{n}(t)$$



Check the Nyquist Criterion

$$g_o(t) = g(t) * g(T_s - t)$$

$$y_m = \sum_{k=-\infty}^{\infty} a_k g_o(mT_s - kT_s) + n_o(mT_s)$$

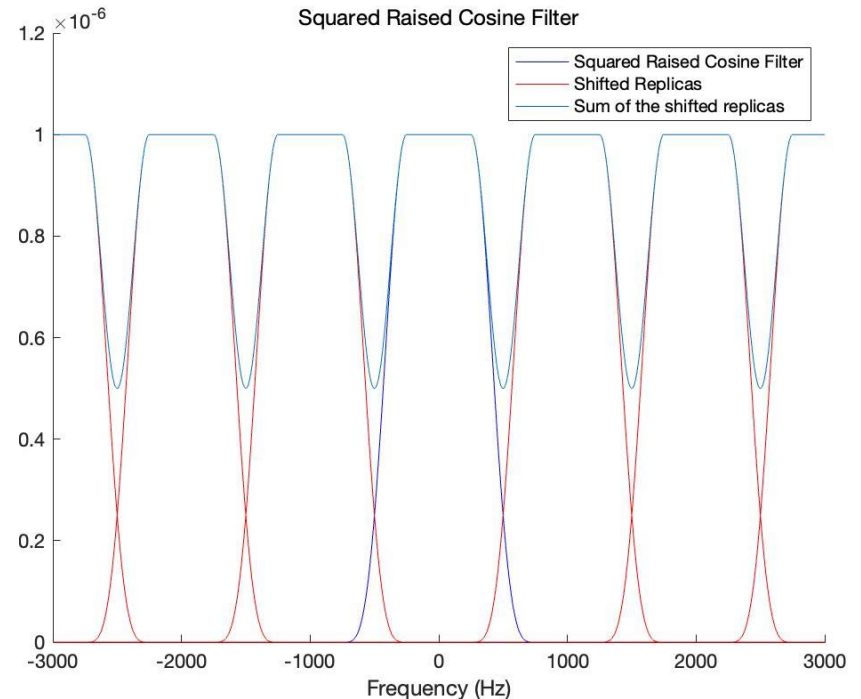
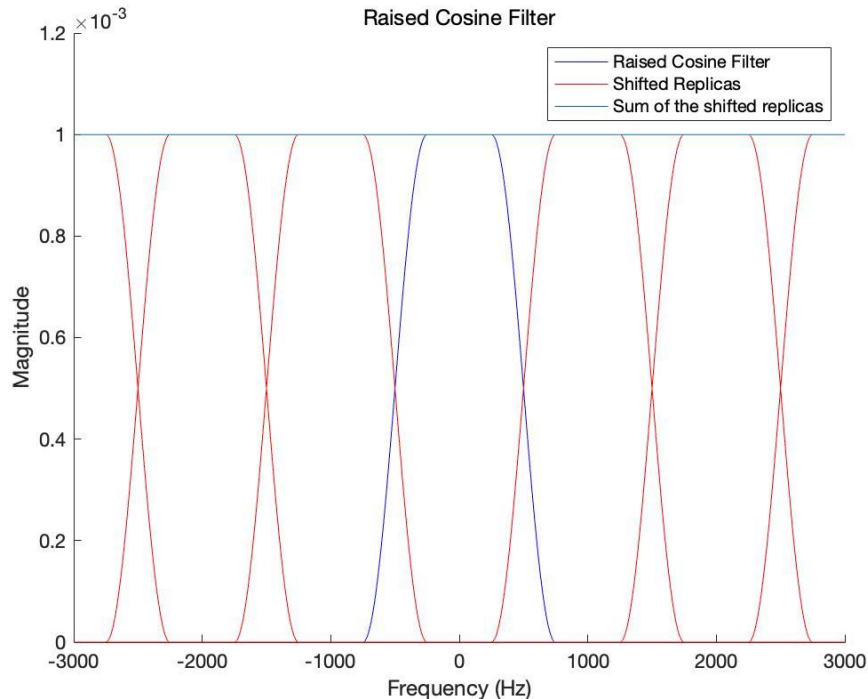
- If $g(t)$ satisfies the Nyquist Criterion, do we have ISI at y_m ?
 - ▶ Suppose $g(t)$ is raised cosine



Check the Nyquist Criterion

$$g_o(t) = g(t) * g(T_s - t)$$

■ Suppose $G(f)$ is raised cosine, then $G_o(f) = |G(f)|^2$



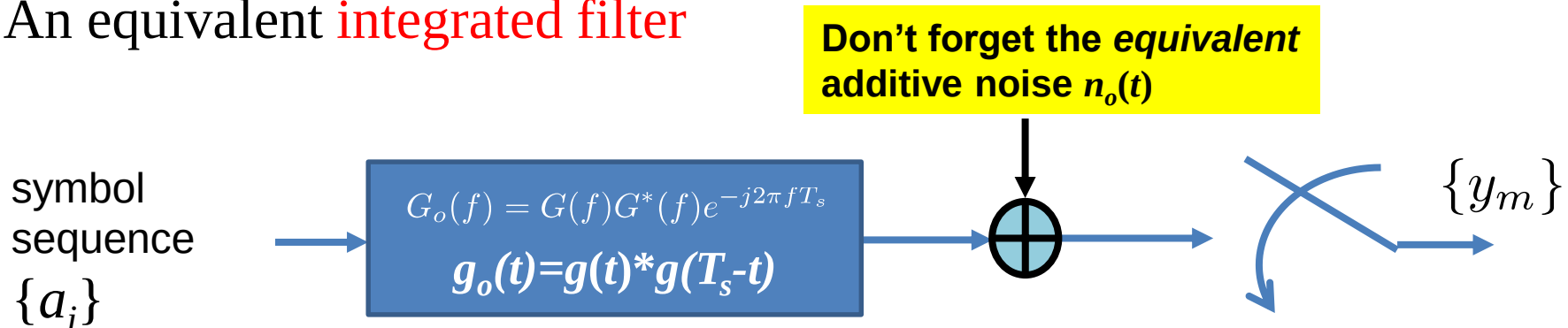


Conceptually Integrate the Transmitter and Receiver Filters



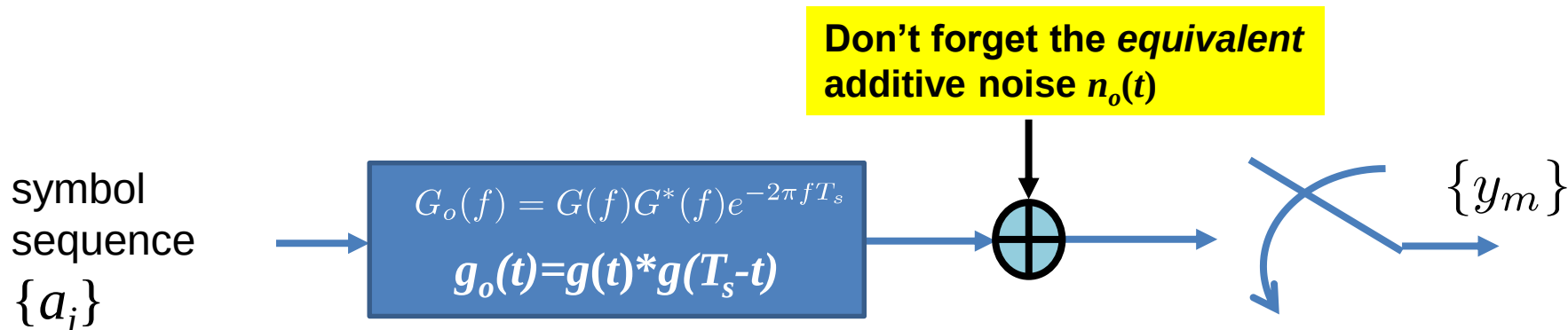
$$y_m = \sum_{k=-\infty}^{\infty} a_k g_o(mT_s - kT_s) + n_o(mT_s)$$

- We only care about ISI at y_m , meaning that $g_o(t)$ should satisfy the Nyquist Criterion
- An equivalent **integrated filter**





Nyquist Criterion on the Integrated Filter



- Recall the Nyquist Criterion is described in the frequency domain
- The following frequency response

$$G_o(f) = |G(f)|^2 e^{-j2\pi f T_s}$$

should satisfy the Nyquist Criterion (here with sample at mT_s for symbol a_{m-1} due to the causality constraint)



Nyquist Criterion on the Integrated Filter

- Therefore we denote some filter satisfying the Nyquist Criterion, say **raised cosine**

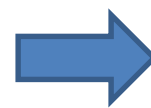
$$G_{\text{rc}}(f) := |G(f)|^2$$

$$g_{\sqrt{\text{rc}}}(t) = \mathcal{F}^{-1}(\sqrt{G_{\text{rc}}(f)})$$

- Therefore, we can let

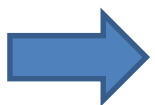
Squared Root Raised Cosine

$$G(f) = \sqrt{G_{\text{rc}}(f)} e^{-j\pi f T_s}$$



$$g(t) = g_{\sqrt{\text{rc}}} \left(t - \frac{T_s}{2} \right)$$

$$G_{\text{m.f.}}(f) = G^*(f) e^{-2j\pi f T_s} = \sqrt{G_{\text{rc}}(f)}^* e^{-j\pi f T_s}$$

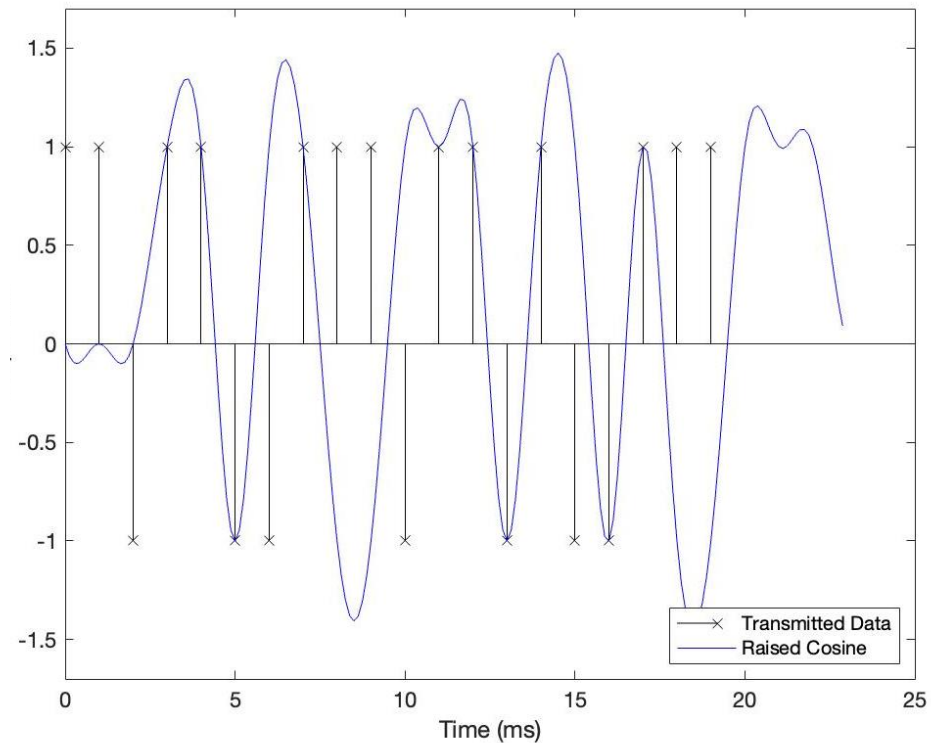
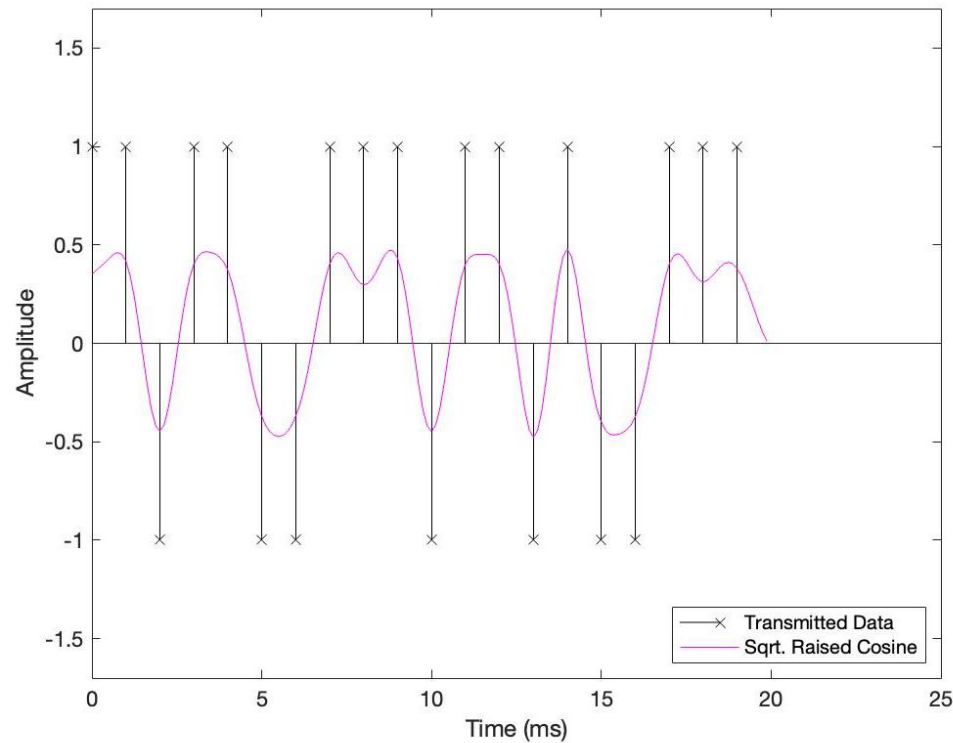


$$g_{\text{m.f.}}(t) = g_{\sqrt{\text{rc}}} \left(\frac{T_s}{2} - t \right) = g(T_s - t)$$



Illustration of Squared Root RC Filter

■ In the time domain:





Outline

- Additive Gaussian Noise
- Matched Filter
- Signal to Noise Ratio



The SNR at the **Output of MF**

$$\text{SNR}_{\max} = \frac{2|a_i|^2}{n_0} \underbrace{\int_{-\infty}^{\infty} |G(f)|^2 df}_{\int_{-\infty}^{\infty} |g(t)|^2 dt}$$

Average SNR

$$\gamma = \frac{2E_s}{n_0}$$

- The waveform of each symbol is $a_i g(t)$
- The energy of this symbol is $|a_i|^2 \int_{-\infty}^{\infty} |g(t)|^2 dt$
- If we average over all possible symbols, the average energy of a symbol, E_s is

$$E_s = \mathbb{E}\{|a_i|^2\} \int_{-\infty}^{\infty} |g(t)|^2 dt = \mathbb{E}\{|a_i|^2\} \int_{-\infty}^{\infty} |G(f)|^2 df$$



E_s/n_0 and E_b/n_0

- We now have the energy per symbol

$$E_s = \mathbb{E}\{|a_i|^2\} \int_{-\infty}^{\infty} |g(t)|^2 dt$$

- What is the energy per bit?

$$M = : |\mathcal{A}| = 2^r$$

$$E_b = \frac{E_s}{\log_2 M}$$

- The nature discretization of digital communications leads to the concept of E_s and E_b even if the analog waveform is used
- E_s/n_0 and E_b/n_0 are very important quantities!



An alternative way of calculating SNR

- From the definition of SNR: signal power versus noise power
 - ▶ Not from the view point of the sample point, but from the entire baseband waveform
 - ▶ Signal power is the energy per unit time
 - Assume that the mean of the symbol $m_a = E[a_i]$ is ZERO
 - Recall the power spectrum density of the baseband waveform

$$S_A(f) = \frac{\sigma_a^2}{T_s} |G(f)|^2$$

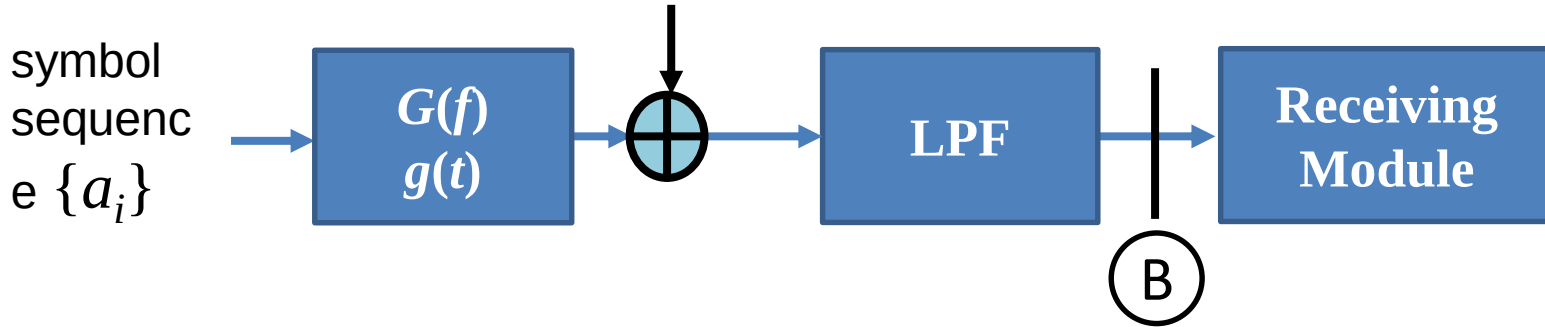
Physical Meaning?

$$S = R(0) = \int_{-\infty}^{\infty} S_A(f) df = \frac{\sigma_a^2}{T_s} \int_{-\infty}^{\infty} |G(f)|^2 df = \frac{E_s}{T_s} = E_s R_s$$



An alternative way of calculating SNR

White Gaussian noise $n(t)$



■ People often put an LPF before the receiving module

▶ The LPF is ideal

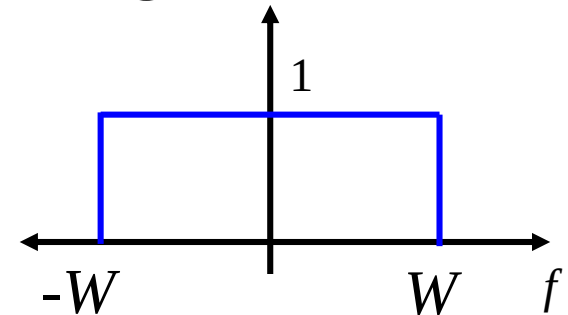
▶ And has the same bandwidth as $G(f)$

■ Noise power at point B: $n_0 W$

■ Signal power is still $E_s R_s$

■ SNR at point B:

$$\gamma_B = \frac{E_s R_s}{n_0 W}$$





Relates Nyquist Criterion with SNR

$$\gamma_B = \frac{E_s R_s}{n_0 W}$$

■ Nyquist criterion indicates that $R_s \leq 2W$

■ The maximum SNR here is

$$\gamma_B = \frac{E_s R_s}{n_0 W} \leq \frac{2E_s}{n_0}$$

■ The equality is achieved with ideal LPF as $G(f)$

■ When we use (squared root) raised cosine as $G(f)$

$$R_s = \frac{2W}{1 + \alpha}$$



$$\gamma_B = \frac{2E_s}{n_0(1 + \alpha)}$$



More about SNR

- However, we previously derived with the matched filter that its output SNR is (for ANY $G(f)$):

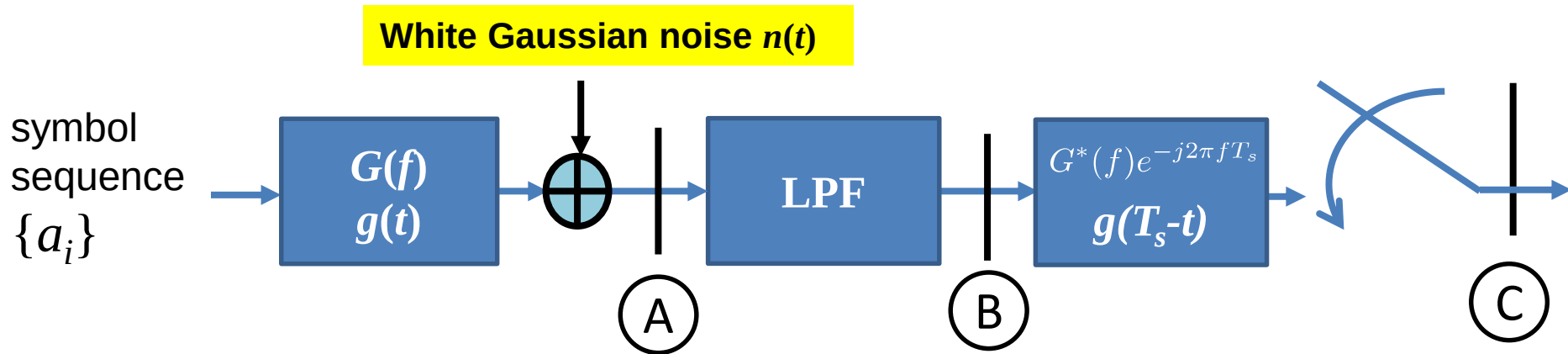
$$\gamma = \frac{2E_s}{n_0}$$

- It is different from: $\gamma_B = \frac{2E_s}{n_0(1 + \alpha)}$

- WHY are they different?



SNR has different values at different points!



- SNR at point A?
- SNR at point B?
- SNR at point C?

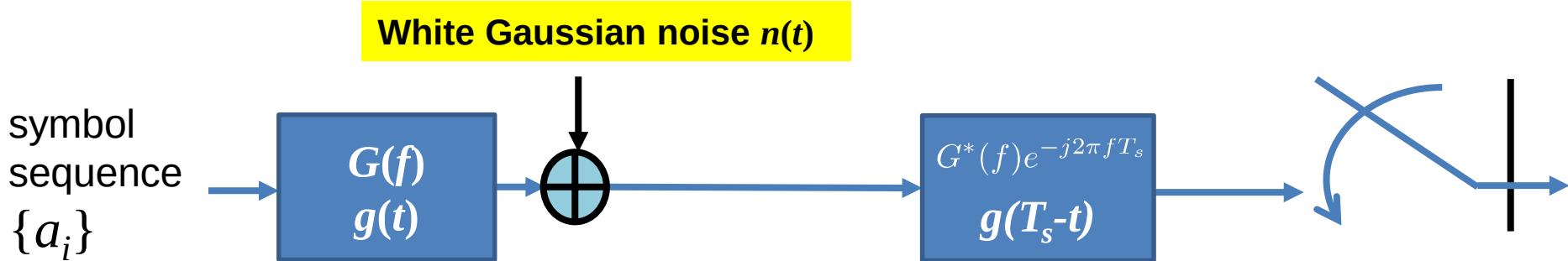
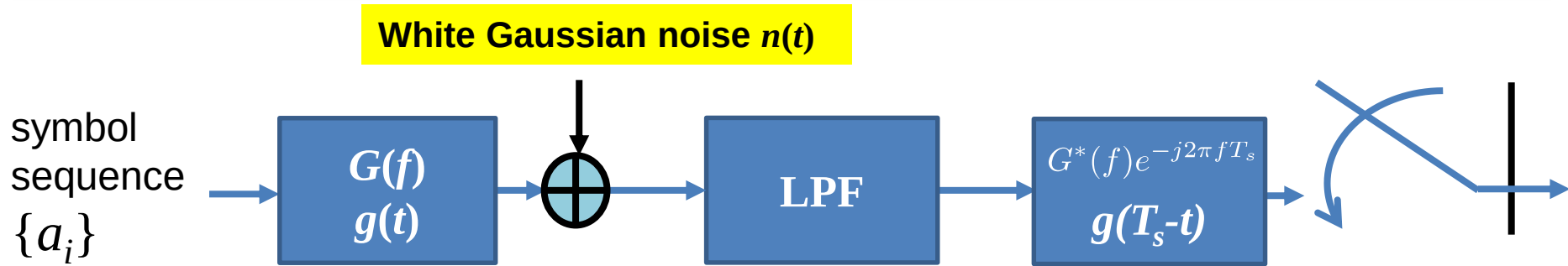
$$\gamma = \frac{\mathbb{E}\{a_i^2\} \left| \int_{-\infty}^{\infty} |G(f)|^2 df \right|^2}{\frac{n_0}{2} \int_{-\infty}^{\infty} |G(f)|^2 df} = \frac{E_s \int_{-\infty}^{\infty} |G(f)|^2 df}{\frac{n_0}{2} \int_{-\infty}^{\infty} |G(f)|^2 df}$$

Larger bandwidth may introduce more noise power, but luckily the signal power is enhanced at the same scale, via the matched filter

Coherent Enhancement!



One more note



■ Are they equivalent??



Summary about SNR

- The SNR of the sample point after the output of the *Matched Filter* is:

$$\gamma = \frac{2E_s}{n_0}$$

- If we use E_b , then

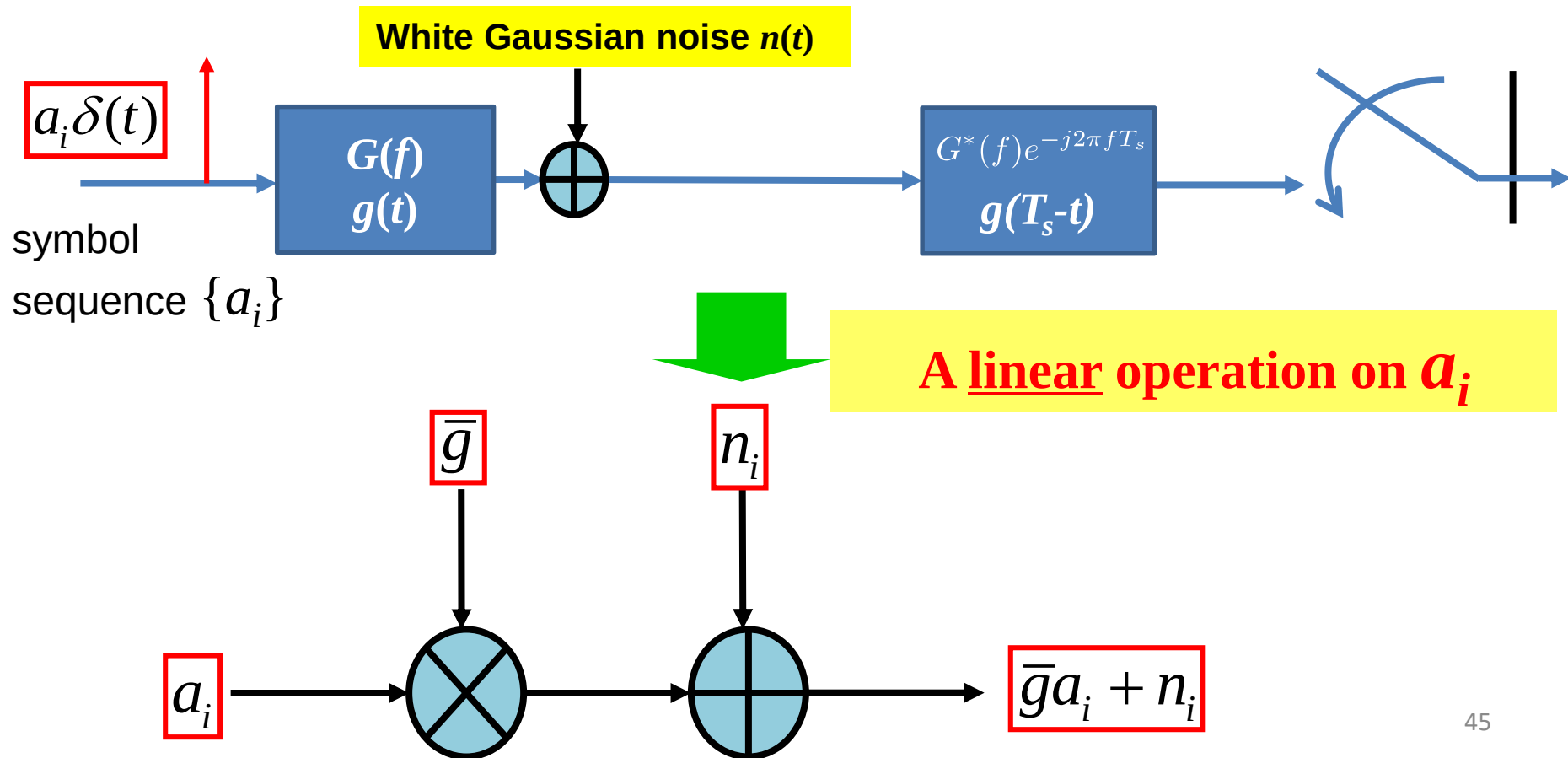
$$\gamma = \frac{2E_b}{n_0} \log_2 M$$

Two most important equations in today's lecture!



Equivalent Baseband Transmission Model

- Through squared root Nyquist waveform filters: **No-ISI**, we can again **focus on each symbol separately**





Equivalent Baseband Transmission Model

- The equivalent linear operation on a_i

$$y_i = \bar{g}a_i + n_i$$

- Remember the SNR at the output of the MF

$$\text{SNR} = \frac{2}{n_0} |a_i|^2 \int_{-\infty}^{\infty} |G(f)|^2 df$$

- And let the noise power unified to $n_0/2$, then

$$\bar{g} = \sqrt{\int_{-\infty}^{\infty} |G(f)|^2 df}$$



Equivalent Baseband Transmission Model

- Keep the SNR, and let

$$y'_i = y_i / \bar{g}$$

- Then

$$y'_i = a_i + n'_i$$

- Define the equivalent noise variance:

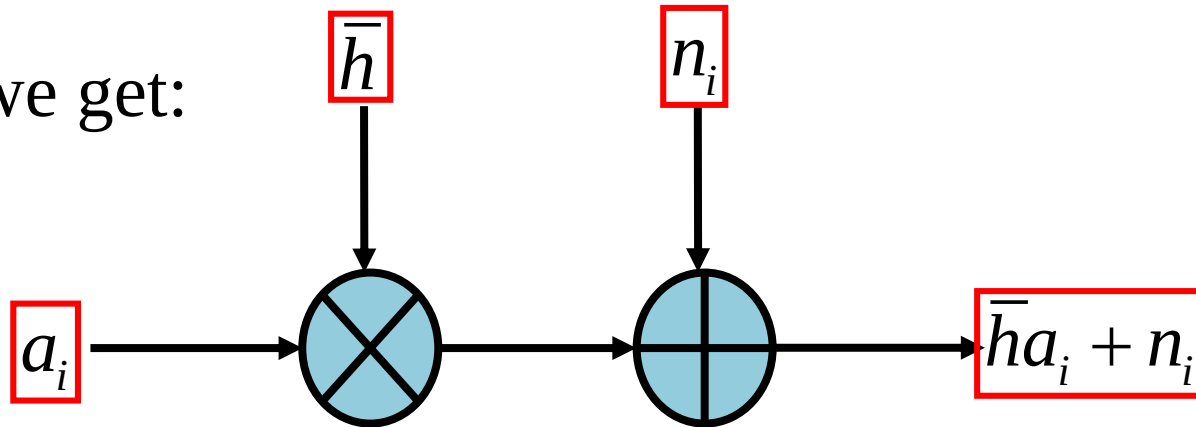
$$n'_i \sim \mathcal{N}(0, \sigma^2)$$

$$\sigma^2 = \frac{n_0}{2 \int_{-\infty}^{\infty} |G(f)|^2 df}$$



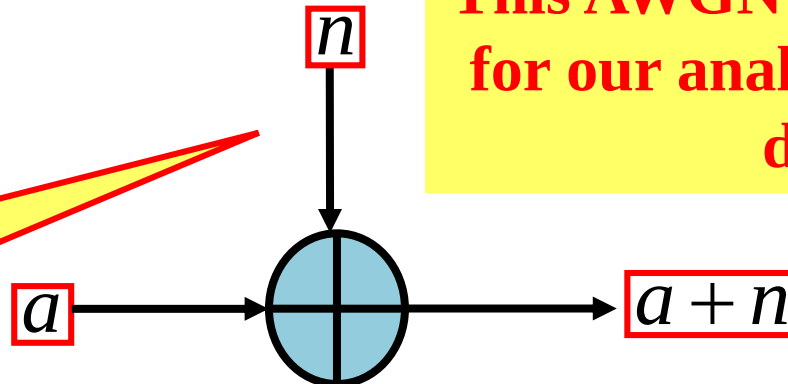
Equivalent Baseband Transmission Model

■ Finally we get:



This AWGN channel is the basis for our analysis for the symbol detection

Coefficients are reflected in the noise power





Summary

- Additive Gaussian Noise
(Haykin - Communication System : 5.9-5.10)

- ▶ PSD density $\frac{n_0}{2}$

- Matched Filter: Maximizes the output SNR
(Haykin - Communication System : 8.2)

$$H(f) = G^*(f)e^{-j2\pi fT_s} \quad h(t) = g(T_s - t)$$

- Signal to Noise Ratio

- ▶ At the output of matched filter

$$\gamma = \frac{2E_s}{n_0} = \frac{2E_b}{n_0} \log_2 M$$